

Wind Blade Inspection System with Unmanned Aerial Vehicle and Ground Robot

by

**Badr Essefiany, Calin Constantin Clichici, Dongwook
Lee, Wail Bougida**

Project Report of Cooperative Work

Supervisor: Acquahmeyer Drone Tech & Prof. Francesco Maurelli

Abstract

The research aims to establish a wind blade inspection system by which an Unmanned Aerial Vehicle (UAV) carries and deploys a ground robot onto the blade surface for contact-based inspection. The work has been split into three subtopics: software control of UAV's flight and positioning, robot arm design for secure ground robot deployment, and optimal ground robot design for vertical movement. For each topic, several methods including already existing approaches and novel ones are presented and tested in the research. For flight and positioning, obstacle avoidance and object tracking algorithms are tested on DJI Tello via Python scripts. For the deployment mechanism using a robot arm, the control mechanism and actuator system material selections are still in comparison based on simulation for the final decision. The design of the mobile robot was accomplished using Fusion 360, different systems were studied and a model with passive suction cups was studied, built and suggested.

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1 Introduction

Recently, maintenance of wind blades has been done mainly by human technicians. Technicians are roped to climb up and check every part of the blade, which is life-risking and also timely and costly expensive. Due to these drawbacks, there have been many efforts to develop human-independent maintenance systems such as UAV's vision-based inspection [20], mobile robot's contact-based inspection [24], and roped robot's repair works [6]. Nevertheless, these approaches are not fully human-independent and not prevalent as well.

Meanwhile, the research proposes a new idea of a novel maintenance system in which the UAV and the mobile robot cooperates. The UAV picks up the mobile robot, flies next to the wind blade, and deploys the mobile robot on the wind blade. The next step of the mobile robot will be further decided based on its objectives, such as inspection and repair work. The overall system structure is visualized in Figure 1.

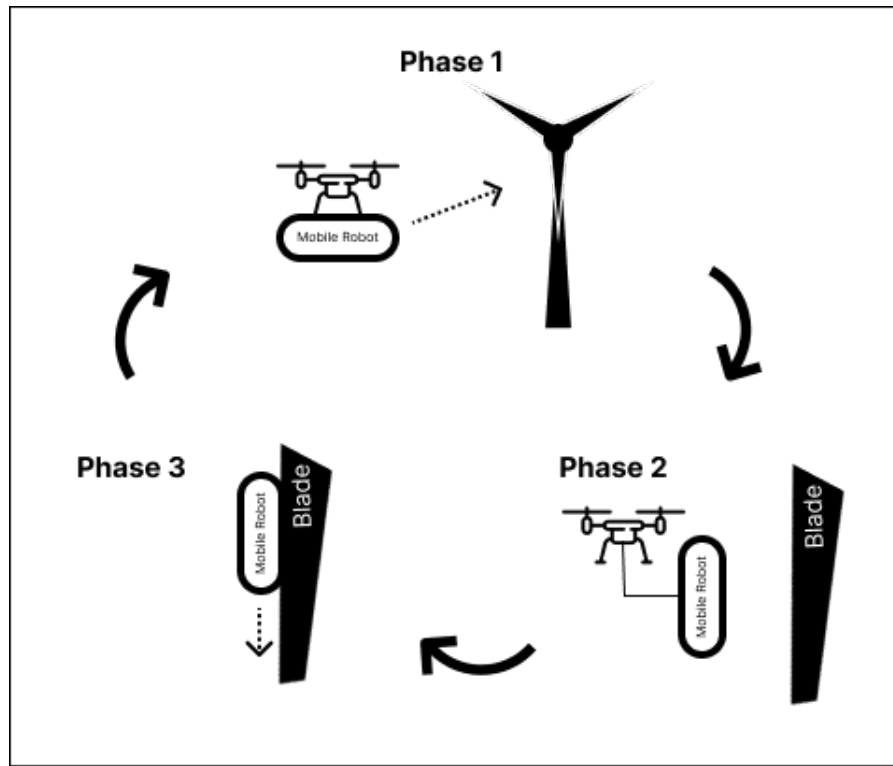


Figure 1: Visualization of the research's proposal system structure

To substantiate this maintenance system, the research is split into three subtopics based on the system phases shown above in Figure 1: UAV's safe flight and precise positioning to the deployment point, UAV's robot arm design and control for secure deployment of the ground robot onto the blade surface, and the ground robot's optimal design for vertical movement. Still, the paper does not cover the control of the UAV's flight and the specific inspection functionalities on top of the ground robot. The control of the UAV is aided by an autopilot suite such as Ardupilot and the ground robot's design only focuses on the vertical movement.

2 Related Works and Literature Review

2.1 Wall climbing robots

Technique	Mechanism	Specification and performance
Suction cups Adhesion	Suction cups depressurized by vacuum generators are commonly used for pneumatic adhesion and adhesion. This strategy is often suited only on smooth climbing surfaces to minimize air leakage.	These types of robots are usually heavy and huge, due to the usage of vacuum generators. A robot with a body length of 25 cm and a usual suction force of 6 N was recorded.[12] A 1.8 m and 45 kg robot with a suction force of 1500 N was recorded. The maximum speed of locomotion is 0.15 m/s. [15]
Magnetic / Electro-Magnetic Adhesion	For climbing locomotion,Permanent magnetization or electrical magnetization are used in this procedure. The permanent magnet solution does not require any energy to remain on vertical walls, and it is frequently extremely strong. However, it may only be applicable to movement on ferromagnetic surfaces.	Even with a small body size, substantial and stable adhesion forces may be obtained.a small-sized robot weighing 2 kg (with a 1.2 kg payload) that can attain 160 N adhesion and travel at a maximum speed of 30 m/min.[11] Another paper demonstrated a 29 cm and 40 kg robot (with a 10 kilogram payload) capable of moving at 8 m/min. [7]
Gripping	Climbing robots use grabbing devices to attach their bodies to gripping surfaces in their environment. Mechanical grippers or unique mechanical connections (relatively small hooks) are typically used on this sort of climbing robot to retain footholds on uneven surfaces with applicable gripping structures.	When hook-type grasping mechanisms are used, fast climbing is feasible. an autonomous climbing robot that is 55 cm long, weighs 5.3 kg (with a 1.5 kg payload), [1] and climbs at a rate of 2.4 m/min was published. Active gripping mechanisms, on the other hand, result in slower but more robust climbing. a 25 cm long and 1.34 kg robot capable of climbing at 0.20 m/min was reported. [25]

Table 1: evaluation of adhesion mechanisms

We conducted a survey about existing climbing robots, Pictures ?? of the robots are located after the table below 2:

Climbing Robot	Adhesion	Locomotion	Movement	Application Task
"Ring" Climb	wheeled	Mechanical spring traction	UP-DOWN rotate	inspection
Magnetic Adhesion	Electroadhesive	Magnetic	UP-Down rotate	inspection/maintenance
OmniWheel	Omnidirectional wheeled	Magnetic	Vertical-Horizontal	inspection/maintenance.
Lego-Based	Tracked	RopesTraction	vertical horizontal	
Bladebug	legged	vaccum	Omnidirectional	inspection/resurfacing.

Table 2: existing climbing robot

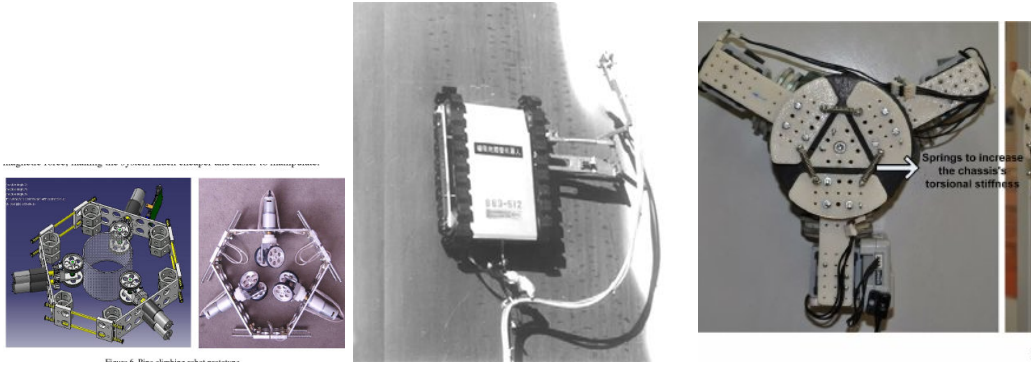


Figure 2: Omniring [21], MagneticTrack [22], Omniclimber [3]

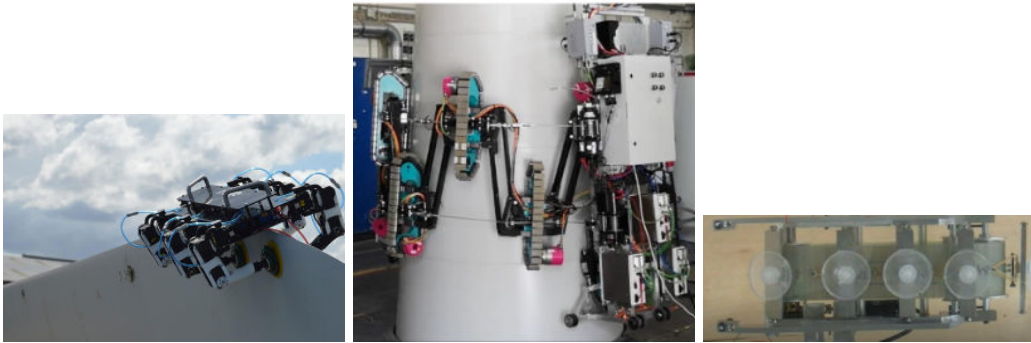


Figure 3: Bladebug [8], Leggobased [23], Suction cups crawlers [9]

2.2 Optical Flow

Optical flow is the vector representing the apparent motion of individual pixels within the visual scene. The pixel movements are understood as the movement of such object, surface, or edge. Optical flow is used in various applications, mainly for motion estimation and object tracking. The research uses optical flow vectors to estimate the motion of the dynamic obstacle invading the camera view of the UAV so that the UAV can further take a pose to avoid the collision. As shown in Figure 4, the high-level understanding of optical flow is done by calculating dx and dy in the current frame image relative to the previous frame image.

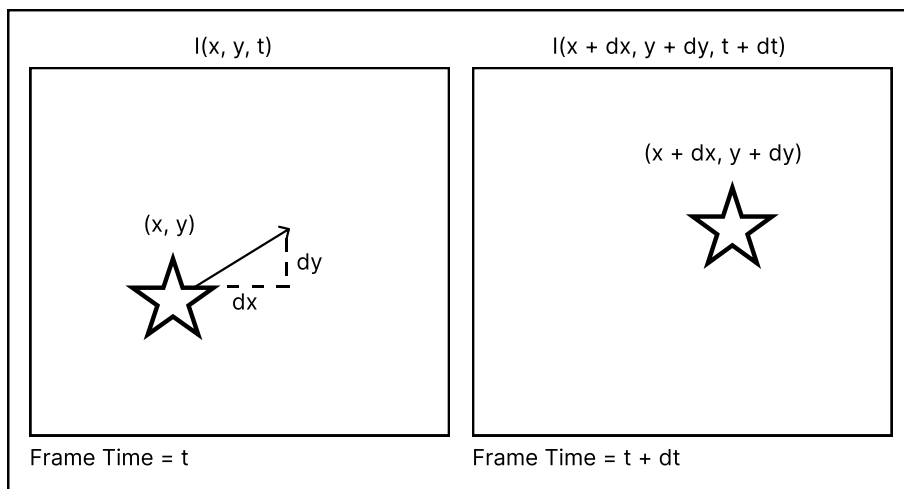


Figure 4: Optical flow's high-level visual explanation

Generally, optical flow algorithms are categorized into two: Sparse and Dense optical flow. Sparse optical flow algorithms calculate the flow vectors of interesting features such as edges or corners of objects, while dense optical flow algorithms calculate the flow vectors of all pixels within the entire frame. Thereby, dense optical flow algorithms have relatively higher accuracy although being relatively slow and computationally expensive [14].

In this paper, one of the dense optical flow algorithms, Gunnar Farneback's algorithm, is used [10]. For this algorithm, the primary assumption is that the pixel intensities are constant between two consecutive frame images. Based on this assumption, the first equation is derived (1), followed by Taylor approximation (2), which leads to (3) with $V_x (= \delta x / \delta t)$ and $V_y (= \delta y / \delta t)$, finally transformed into matrix form in equation (4) and (5). I_x , I_y , and I_t stand for the derivation of pixel intensity on the subscripted axis.

$$I(x, y, t) = I(x + dx, y + dy, t + dt) \quad (1)$$

$$I_x \delta x + I_y \delta y + I_t \delta t = 0 \quad (2)$$

$$I_x V_x + I_y V_y + I_t = 0 \quad (3)$$

$$I_x I_y V_x V_y = -I_t \quad (4)$$

$$V_x V_y = I_x^2 I_x I_y I_y I_x I_y^2 - I_x I_t - I_y I_t \quad (5)$$

2.3 Scale Invariant Feature Transform (SIFT)

SIFT is one of the classical computer vision techniques in which the keypoints of the reference image are detected and corresponding descriptors are calculated to be compared with other test images later. At the point of detection later, the descriptors of keypoints within the test image are calculated and matched with those of the reference image. Then, the good matches are filtered after matching the keypoints based on each descriptor [18].

The keypoints detection within the image is done by calculating the Difference of Gaussians (DoG). DoG helps define the outstanding points within the image by subtracting several layers of blurred images. When DoG finds the keypoints, the feature descriptor of each keypoint is to be the Histogram of oriented Gradients (HoG). By comparing the descriptors of keypoints of the reference image with those of the test image, the prior matching is done, followed by the filter of good matches. The filter of good matches is done by Hough Transform. Three or more keypoints that agree on the specific pose of a shape created by Hough transform are clustered to be good matches. Figure 5 shows the result of applying SIFT on a test image with a QR code as a reference object.

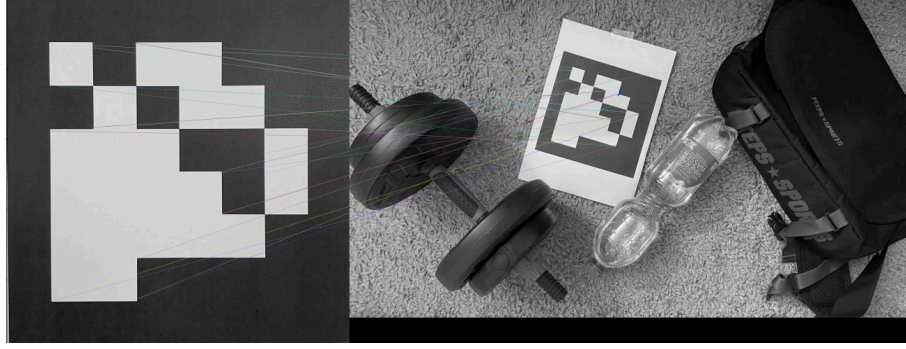


Figure 5: SIFT test on sample reference image (left) and test image (right)

3 System Structure

3.1 Phase 1: Drone Approaches the Wind Blade

The overall flight path of the UAV to the wind blade has three sub-paths. Path 1 is the path from starting point to WP1, Phase 2 from WP1 to WP2, and Phase 3 from WP2 to the goal position WP3, as depicted in Figure 6.

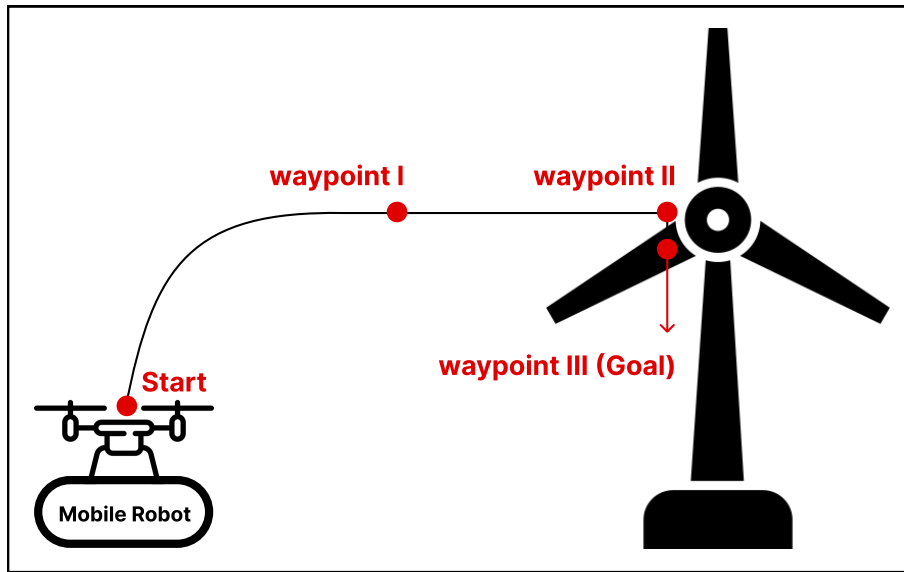


Figure 6: Waypoints of the UAV's overall flight to the wind blade

3.1.1 Sub-Path 1: Obstacle Avoidance Flight

In sub-path 1, while the UAV is controlled by autopilot software in waypoint flight mode, the UAV is vulnerable to collisions with unexpected dynamic obstacles. Thereby, the settlement of efficient obstacle avoidance is required. Among various possible approaches, the research focuses on achieving the goal of obstacle avoidance solely based on the monocular camera view image [5]. Calculation of optical flow vectors is tested to achieve the obstacle avoidance.

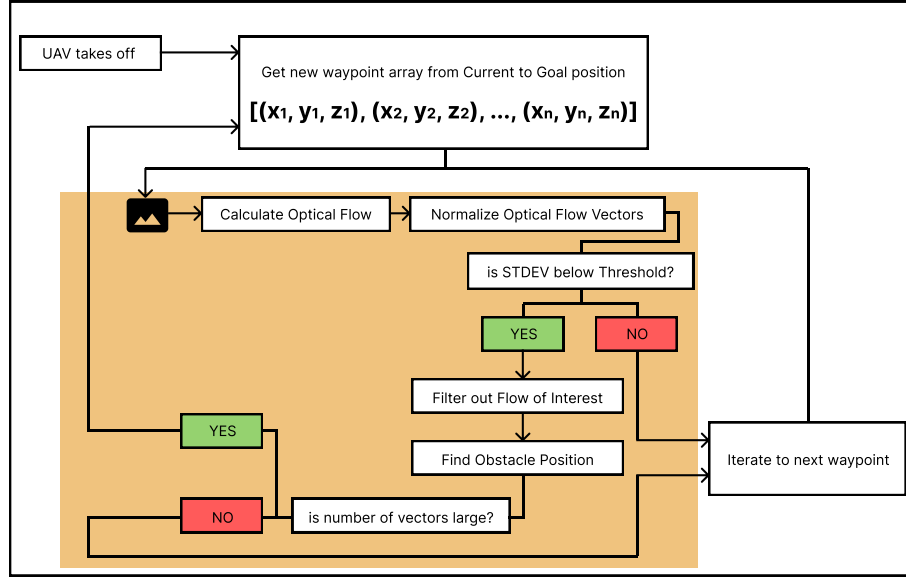


Figure 7: UAV's optical flow based obstacle avoidance system

I. Get waypoints to the goal position

Initially, the array of every specific sub-waypoints from the start point to WP1 is calculated. Then to ensure the UAV's safety, we analyze the view of the UAV every frame to figure out if any dynamic obstacle is detected. If yes, the UAV behaves to avoid the obstacle prior, and if not, the UAV keeps its way onto the next waypoint. In reality, however, we do not have to know the positions of all sub-waypoints; instead, we perform obstacle detection on every frame image in the background while the UAV iteratively follows the path in the waypoint flight mode of autopilot software. At the point where the UAV performs the avoidance action, the autopilot software will re-calculate the path to W1.

II. Calculate Optical Flow

The calculation of optical flow vectors is developed in Python with the help of a well-known computer vision package, 'OpenCV-Python(OpenCV)', where Gunnar Farneback's algorithm is pre-implemented as a function. In addition, the pyramidal resolution change of the image is included in the function, in which several layers of different resolution images are being processed with Gunnar Farneback's algorithm separately. The objective is that since the optical flow vectors are calculated related to each pixel's neighbors, the large pixel displacements are expected to be found easier in the reduced-sized image and small changes in the expanded-sized image. Then, all the optical flow vectors found in several layers of the image are integrated as a result. After calculating optical flow vectors, the computation goes through additional processes for proper analysis of the optical flow vectors and detection of threatening dynamic obstacles.

III. Get Frame View Activeness

To categorize the status of the frame view into active or static, the standard deviation(STDEV) of the overall optical flow vector magnitudes is computed. Since optical flow vectors are sensitive to the minor changes in pixel and thereby tend to include the noises in every flow vector, the STDEV value is a proper measurement that can notify if the entire flow vectors

are about the same value. By setting the lower-bound threshold, we regard the frame view as active if its STDEV is larger than the threshold.

However, the primary concern of using the raw value of optical flow vectors is that the relative velocity of the static instance within the view has to be ignored. The research attempts to exclude the relative velocities by normalizing the flow vectors. For an easier manual decision on the thresholds, the method scales up the normalized flow vectors into a specific numerical range (0, T1) by multiplying each flow vector with T1. In the test, T1 is set as 150. Then, the frame is regarded as active if the STDEV of entire flow vectors is larger than a user-defined threshold(T2). Figure 8 shows the example of an active frame being alerted by the red circular mark appearing on the left top of the image.

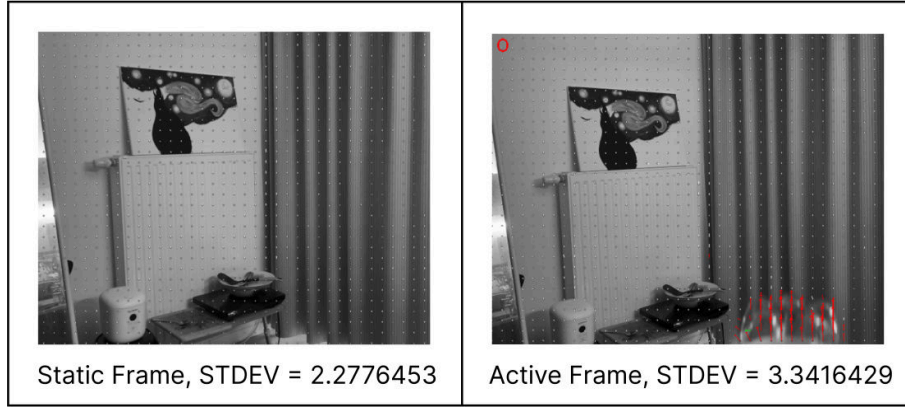


Figure 8: Static(left) and Active(right) frame. (T1 = 150, T2 = 3)

IV. Get Flows of Interest

When the frame is active, it leads to the step of finding and grouping the optical flow vectors of interest. Two factors of the flow vectors are considered: magnitude and direction. The first step is to perform non-maximum suppression. The threshold used for the suppression step is T3, which is set to be 0.7 in the research. The maximum magnitude value among the flow vectors of a specific frame image is multiplied by T3; all the other flow vectors with a magnitude equal to or larger than the value are considered to be of interest. The flow vectors that survived from non-maximum suppression are called 'flow vectors of initial interest'.

After that, the flow vectors' direction is examined. Given a dynamic obstacle in the view, the UAV does not have to waste its power to avoid the obstacle if the obstacle is not moving toward the UAV. According to each flow vector's direction, the UAV can tell if the flow vectors of initial interest are directed toward the center or not. Given the center's position $C = (c_x, c_y)$, the euclidean distance from each flow vector's starting point $P_s = (x_s, y_s)$ and terminal point $P_t = (x_t, y_t)$ to C is calculated. Each distance is represented as D_s and D_t in equation (6).

$$D_s = \sqrt{(x_s - c_x)^2 + (y_s - c_y)^2} \quad D_t = \sqrt{(x_t - c_x)^2 + (y_t - c_y)^2} \quad (6)$$

Then, in the case where D_s is larger than D_t , the flow vector is heading center. Finally, the flow vectors heading center are sub-grouped, called 'flow vectors of most interest'. As depicted in Figure 9, the test result of calculating the optical flow vector is shown. Normal flow vectors

are colored white, flow vectors of initial interest are colored green, and flow vectors of most interest are colored red.

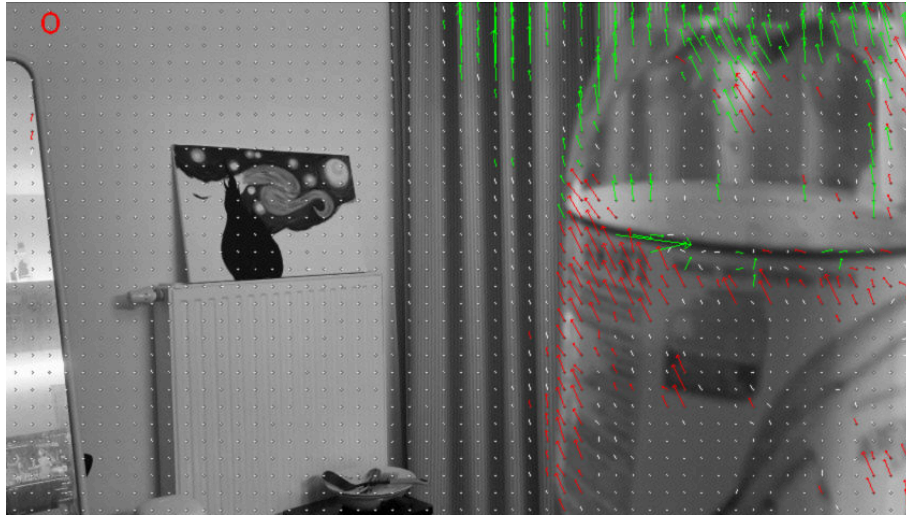


Figure 9: Sample frame with moving water bottle. ($T3 = 0.1$)

V. UAV Pose Determination

To determine the following avoidance behavior of the UAV, the method checks if more than a specific number of threatening vectors are detected. To avoid the cases of noise being detected as a threat, this step of the method also needs to check if the threatening vectors lie in such a bounded area. Even if several threatening vectors are detected in the entire view, the case is ignored if the vectors are spread around. The image view is segmented into five parts for the procedure, as shown in Figure 10 [17].

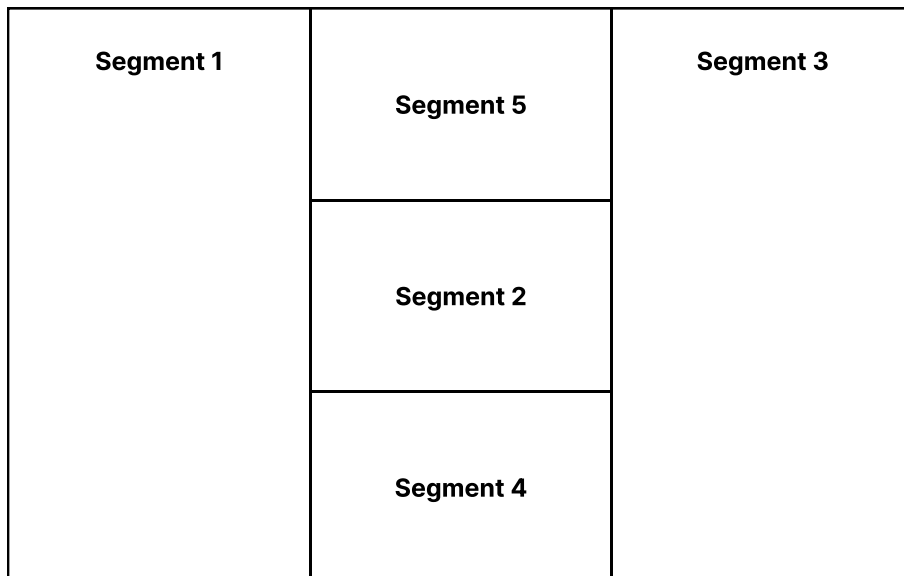


Figure 10: UAV's camera view with split segments

According to which part of the view includes how many threatening vectors, the UAV's decision on where to move for avoidance is done. To avoid the dynamic obstacle, the UAV can move to

an opposite position from where the threat is detected. The UAV will move in a commanded direction with high velocity for immediate and ensured avoidance. More to take into account is the power consumption of the UAV. Based on the fact that the UAV requires more power to move up and down, the research set the UAV's avoidance movement direction to left and right only. When enough threatening vectors are found in segment 1, the methods command the UAV to move right, and when segment 2, 3, 4, and 5 detects threats, the method commands the UAV to move left. By checking the number of threatening vectors within each segment, the method can check whether the threatening vectors are clustered, not ignored as noise.

3.1.2 Sub-Path 2: Object Tracking

Object tracking is for the UAV to move with accuracy to arrive at the wind hub with accuracy. The research proposes a simple approach of the UAV tracking a unique QR code, which is also used in other fields such as for the mobile robots to find their right place at the docking station [4]. The QR code will be pasted or painted at the wind turbine's hub center. Figure 11 shows the entire process diagram of QR code detection and the following pose determination of the UAV.

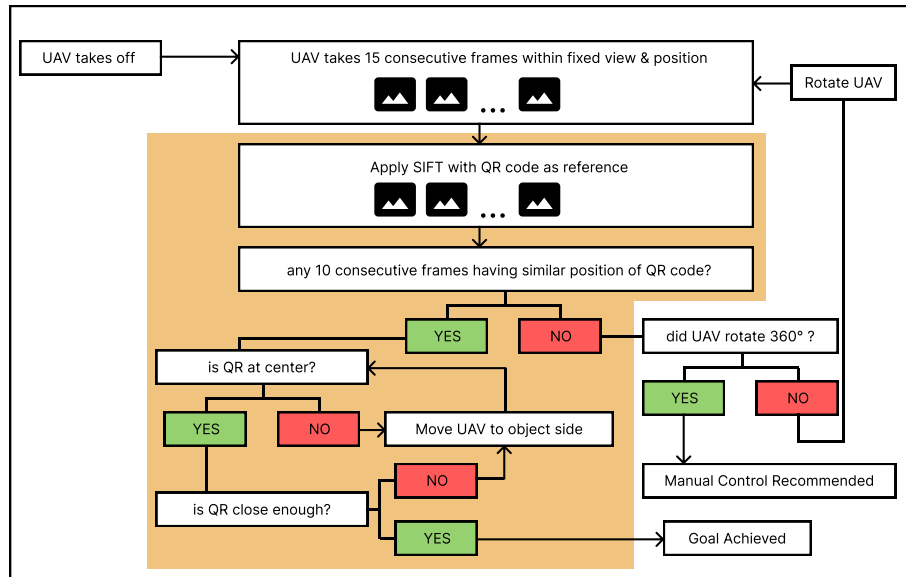


Figure 11: UAV's object tracking system diagram

I. Detection of QR code

To track the QR code, the detection comes in prior. For the detection, the research used SIFT method. As explained above in related work, SIFT method provides the matching results of the descriptors of similarity. The research, in addition, has elaborated the visualization of the detected object for a more straightforward representation and further analysis. Matching lines are eliminated, while the bounding box, center, and bisection line are added. The bounding box is created by finding the homography matrix of good SIFT matches. Figure 12 is a snapshot frame image of a real-time object detection test done on a webcam.

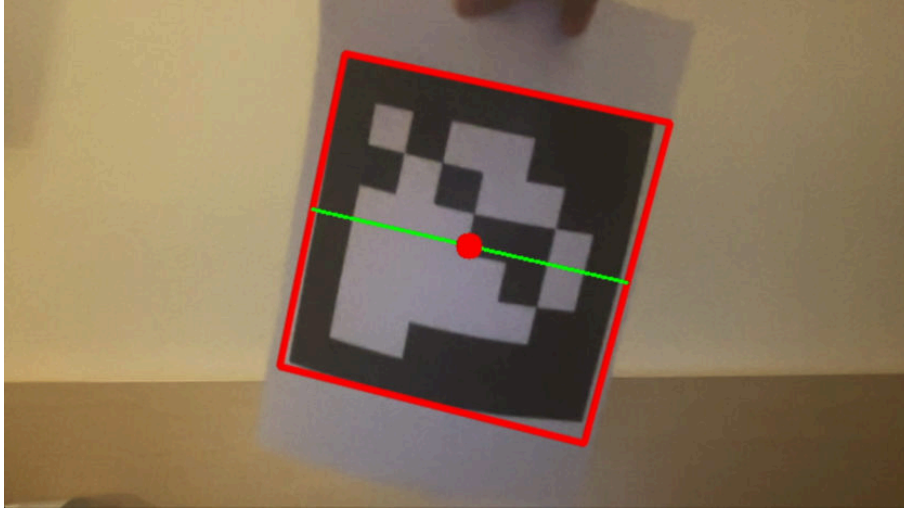


Figure 12: Detection result of QR code based on SIFT

However, it is hard to tell if the detection result is true by only observing a single frame. The reason is that since SIFT matches the descriptors of similarity, there can be cases of false matches on other feature points. Still, as an advantage, the UAV's frame view is not perfectly static when it is hovering due to the turbulence; that false SIFT result on wrong feature points will change instantly frame by frame. Since it also means that the SIFT result will not remain the same in consecutive frames if the match was false, we can filter the true cases by comparing the specific number of consecutive frames. Figure 13 represents the case of three consecutive frame images with false SIFT results. We can see that the center position of the bounding box of which SIFT is regarded as an object in each frame differs a lot.

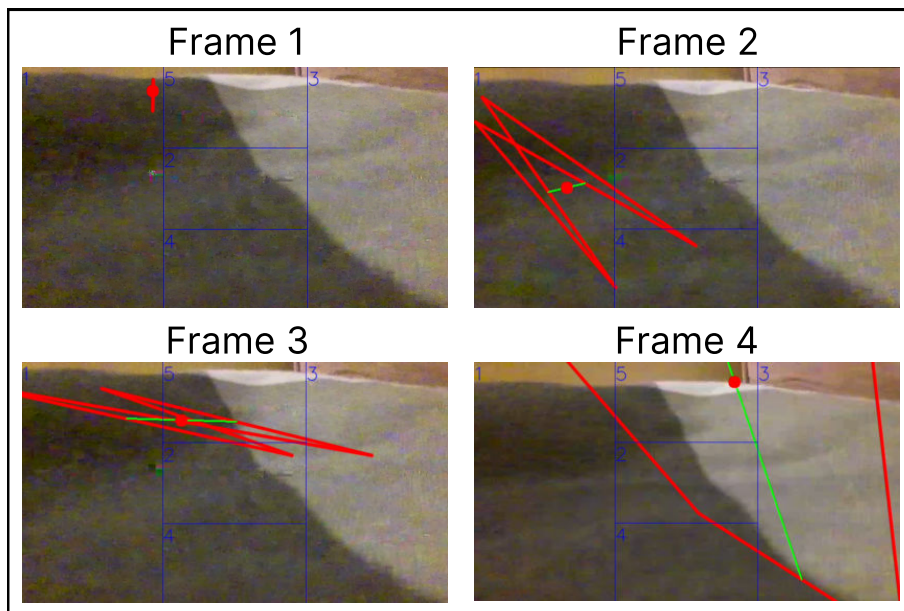


Figure 13: Consecutive frames of False SIFT results

Due to this aspect, the simple filtering of whether the detection is valid or not is done by checking the specific number of consecutive frames. The research expects 10 out of 15 con-

secutive frames to be detecting the QR code in a similar position, within 50 pixels difference. The number of frames to check and allowed error margin pixels can be customized.

II. UAV Pose Determination

Based on the true detection result of SIFT on the QR code, the UAV checks its position. The method uses the same view segmentation as Sub-Path 1, Figure 10. Differently, the object tracking method commands the UAV to move in the direction where the object is detected. For precise movement, the power consumption comes posterior, that the UAV will also move up and down in Phase 2. Depending on the center position of the bounding box, which is depicted as a red dot in the middle of Figure 12, the UAV will move.

In the cases where QR positions on the left, right, bottom, and top sides of the UAV's view, the UAV moves in the corresponding direction. Then, when QR lies at the center, the UAV moves forward until it gets close enough. As there is no distance measurement unit on the UAV, the estimation is done by getting the pixel width of the bisection segment of the bounding box. The setting for the test is to stop the UAV when the bisection segment's length in pixels is over 50 percent of the camera view's width in pixels. For example, a 1280 × 720(720p) resolution video will regard the QR code close enough if the width of the QR code is found to be over 640 px. The UAV is commanded to land after a close-enough approach in the test. Later in the actual launching phase of the project, however, the UAV will move onto its next phase of mobile robot deployment when it arrives at the point.

3.1.3 Sub-Path 3: Descent of the UAV to the top of single wind blade

After the UAV arrives in front of the wind turbine's hub using object tracking, it is expected to descend itself down and position it near the top surface of a single wind blade. The UAV will land the mobile robot at the position and let it start the next phase of contact-based maintenance. The distance of how much to lower the UAV will be decided by the system variable set beforehand: the wind turbine's hub radius.

3.2 Phase 2: Robot Arm deploys the Mobile Robot

3.3 Phase 3: Mobile Robot Design for Vertical Inspection

3.3.1 User requirements

User requirements are those established by the end user. These specifications specify how a facility, piece of equipment, or process should perform in terms of the product to be created, needed throughput, and manufacturing circumstances. User requirements give information that may be used to further design, develop, and test a manufacturing system. In our case, the user requirements are as follows.

- Weight limit of 15kg.
- Dimensions limit
- Weight distribution and position of center of mass.
- Usage of proper adhesive material; high adhesive and non destructive to the blade surface.
- Adaptability to terrain changes.
- Type of locomotion.

- Low Energy Consumption.
- Safe and reliable design.

3.3.2 Subsystems

The belt-pulley system provides a linear non-periodic function of mechanical energy transmission.

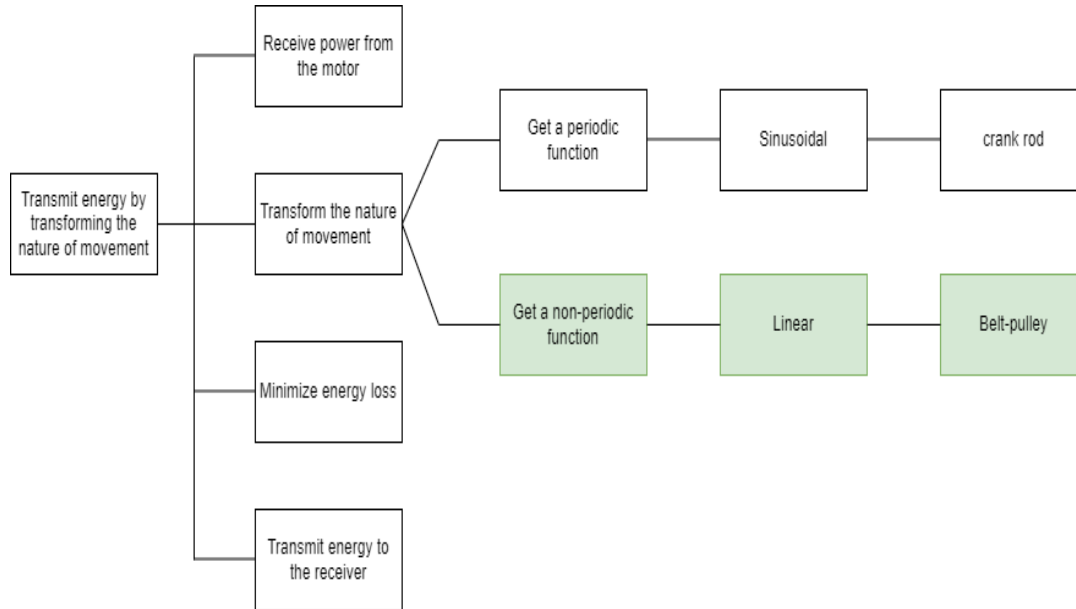


Figure 14: Requirements

Suction cups The scale used to identify the hardness of suction cup materials is Durometer Scale A, which ranges from hard to soft. there are many materials used in the suction cup manufacturing such as silicone has an extreme temperature resistance high/cold but is able to damage the surface; polyvinyl chloride(PVC) is a low cost material with a limited temperature range and durometer, offering a short lifespan. NBR has long lifespan two time silicone life span and cost 30than silicone; Polyurethane has a high wear resistance and is very expensive; HT1 has a very high temperature resistance and do not damage surface plus having longer lifespan then all other materials. The holding force for one suction cup can be expressed as:

$$F = \Delta P \times A$$

- F : holding force in N
- ΔP : difference between ambient pressure and pressure of the system
- A : effective suction area

While for a vertical vacuum cup that moves a weight vertically,the forces can be expressed as:

$$F_{th} = (m/\mu) \times (g + a) \times S$$

where:

- F : holding force in N

- m : lifted mass in kg
- μ : friction coefficient
- g : acceleration of gravity in m/s^2
- S : safety factor

Tracked Belt System The purpose of the tracked belt is to Transmit power between two generally parallel distant shafts by adhesion.

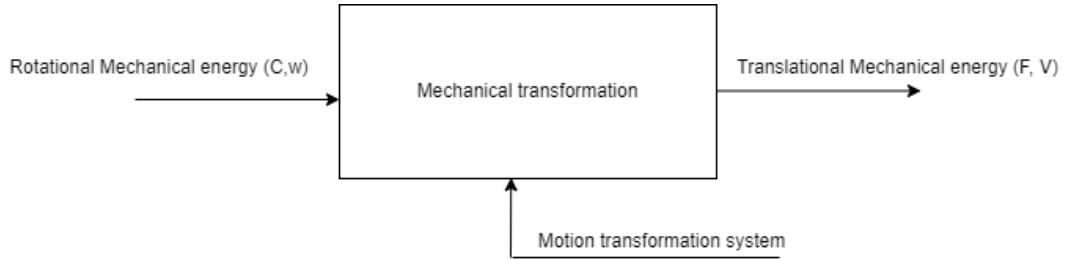


Figure 15: Energy transformation

Kinematic diagram The kinematics of the diagram is as follows :

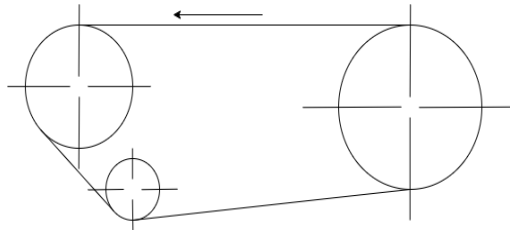


Figure 16: Kinematic scheme

Kinematic calculations The equations translating motion in this systems are:

$$V = \frac{N_d p \cdot Z_d}{60} \qquad k = \frac{\omega_D}{\omega_d} = \frac{d_p}{D_p} = \eta \frac{C_D}{C_d}$$

Where:

- N_D : speed of the large pulley in rpm
- N_d : speed of the small pulley in rpm
- ω_D and ω_d : speeds in rad/s
- d : small pulley winding diameter
- D : large pulley winding diameter
- C_D : torque on the large pulley in N.m
- C_d : torque on the small pulley in N.m

The motor torque required for constant velocity in a belt drive system is based on the following equation:

$$T_c = \frac{m \cdot g \cdot \mu \cdot r_1}{1000 \cdot \beta}$$

Where:

- m : mass of moved load in kg
- g : gravity m/s^2
- μ : coefficient of friction of guide
- r_1 : radius of drive pulley in mm
- β : efficiency of belt drive system

For acceleration and deceleration phases, the equation are the following:

$$T_{acc} = T_c + \alpha \times (J_m + J_c + J_{p1} + J_{p2} + J_l)$$

$$T_d = T_c - \alpha \times (J_m + J_c + J_{p1} + J_{p2} + J_l)$$

Where: !All inertia are in kgm^2

- α : angular acceleration in rad/s^2
- J_m : inertia of mass
- J_c : inertia of coupling
- J_{p1} : inertia of drive pulley
- J_{p2} : inertia of driven pulley
- J_l : inertia of load

Incline angle α An angle was introduced to smooth the connection.

The incline angle α aims to ensure a smooth connection with the surface, in the absence of this angle, the edge of the suction cup will touch the surface first, which will result in the deformation of suction cup, which leads to loss of adhesion force.

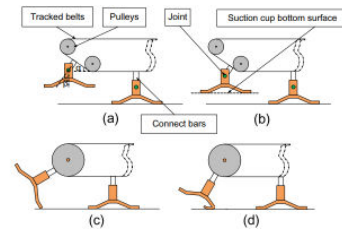


Figure 17: incline angle [9]

Belt Belt transmission systems provide a quiet and smooth operation, good grip, and are suitable for transmitting high power; however, slippage appears to occur, resulting in a non-regular gear ratio; this problem is corrected by a poly V belt, which has excellent adhesion and allows the transmission of significant powers. The V-belt is used to increase the contact area while decreasing the radial force, allowing it to transmit more power than flat belts. Furthermore, numerous belts must be used in parallel on the same pulley to convey large power. One of the pulleys must be fitted with flanges in order to prevent axial slippage of the belt. They can be thought of as flat belts with teeth. They work by meshing (without slipping) like a chain but with more flexibility. Unlike other belts, they handle low speeds well

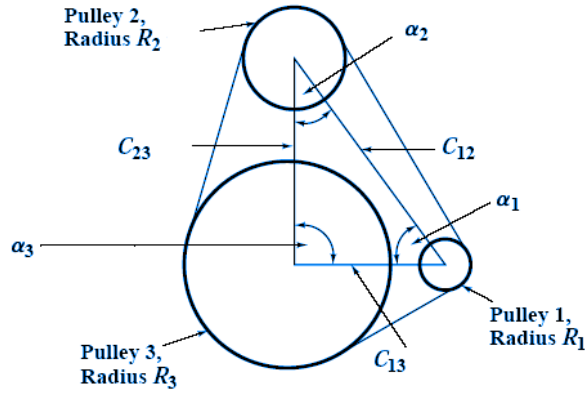


Figure 18: three pulleys system

and require lower initial tension. The length of a belt traversing three pulleys, as shown in figure above, can be found by the following equation:

$$L = c_{12} + c_{13} + c_{23} + \frac{1}{2} \cdot \left(\frac{(R_2 - R_1)s}{c_{12}} + \frac{(R_3 - R_1)s}{c_{13}} + \frac{(R_3 - R_2)s}{c_{23}} \right) + \pi(R_1 + R_2 + R_3) - (\alpha_1 R_1 + \alpha_2 R_2 + \alpha_3 R_3)$$

Where:

- $\alpha_1, \alpha_2, \alpha_3$: degrees in rad
- R_1, R_2, R_3 : radius in mm
- C_{12}, C_{13}, C_{23} : lengths in mm

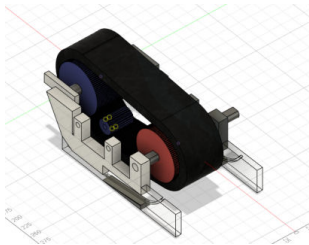


Figure 19: 3D model

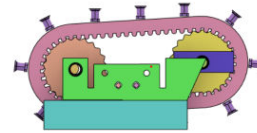


Figure 20: 2D Side View

3D model

Guide Rail The guide rail has the function of:

- Equal load distribution on each suction cup.
- Make the first suction cup contact the blade smoothly.
- Make contact between suction cup and blade almost perpendicular.
- Press the front suction cup while drawing the rest connected suction cups in the pulling area.
- Transfer and nullify the reaction forces caused by pressing the suction cups to the surface.

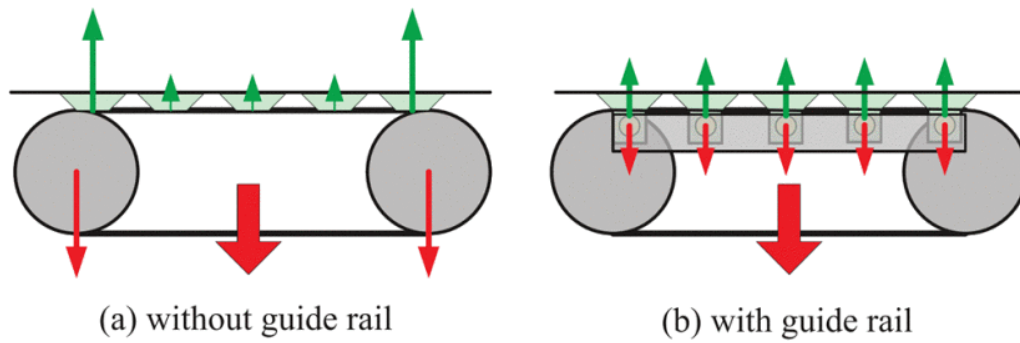


Figure 21: Purpose of Guide rail [27]

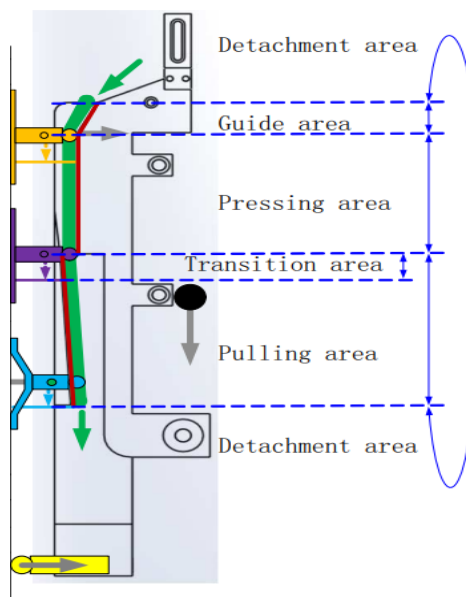


Figure 22: Areas of motion

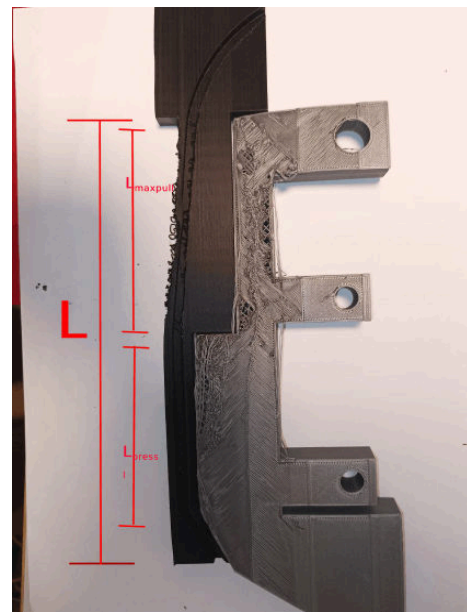


Figure 23: 3D Printed Guide-rail

Areas of motion The area of motion could be seen as:

We can divide the thing to 5 main area:

Area	Description
Guide	In front of the robot, an inclination angle α is intended to ensure a smooth connection to the surface.
Pressing	The guide rail provides a sufficiently strong pushing force to the suction cup.
Transition	The guide rail pulls up on the back attached suction cup to regulate its adhesive force.
Pulling	Not found
Detachment	the suction cup detaches from the surface.

Table 3: Guide rail Design

Evaluation Evaluating our results.

I guide	I press	I max pull	H max pull	H guide
32 mm	81 mm	100 mm	10 mm	15 mm

Figure 24: Guide rail specifications

3D design model To supply response forces from the surface, a tail attached to the guide rail is used. The tail is a required components for every wall climbing robot, it has as a functionality to balance gravity and drive the robot head to come near to the wall adhering suction cups through the guide rail.

You can see attached a picture of the guide rail

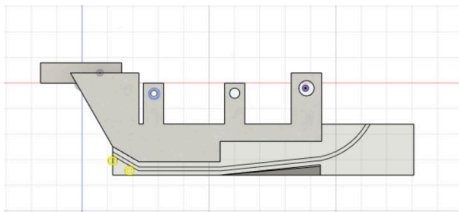


Figure 25: 2D side view of guide Rail model

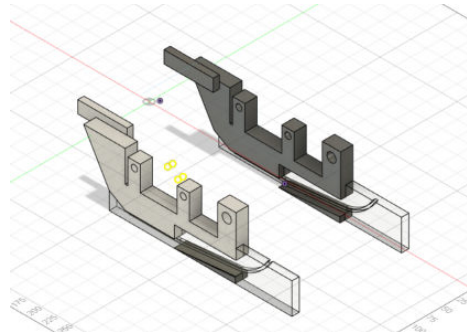


Figure 26: Guide Rail 3D Design

Rotation chassis and Swivel In order to achieve the turning motion, we have implemented a system with two straight-moving modules, we introduced a principle of differential driving where the modules climb up under different velocities.

However, suction cups may not move relative to the belt, and thus a lateral force can not be produced. To solve this problem, we introduce a swivel to ensure the robot turning ability, the modules are connected with each-other through a swivel connected to the payload. This idea creates a multi-body robot capable of wall climbing and improves maneuverability in a given surface.

3D model Here attached is the 3D model

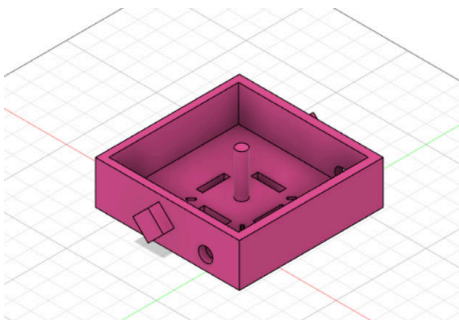


Figure 29: Payload 3D model

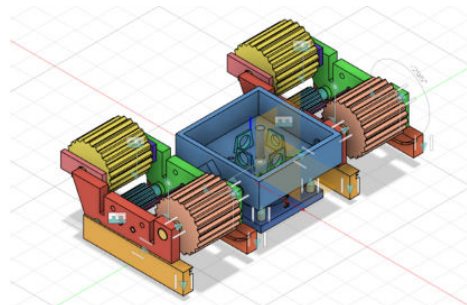


Figure 30: 3D model of swivel with subsystems attached

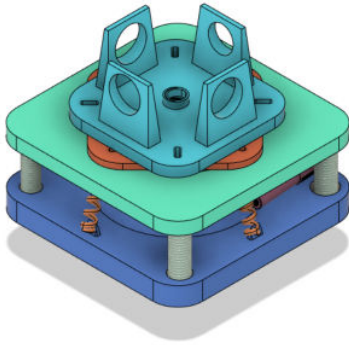


Figure 27: swivel 3D model

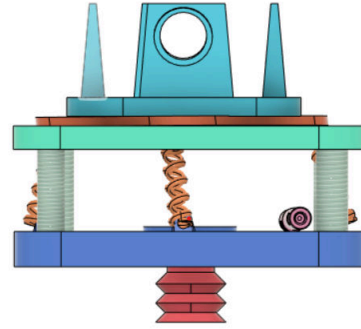


Figure 28: 2D side view

3.4 Robot assembly

Linear locomotion of the system is driven by 2 DC brushed motor (12V Low Noise DC Motor 146RPM with Encoder). A force sensor (joy-it module, SEN-HX711-20, HX711, with 20kg load) is used to record the external payload force and validate the choice of suction cups. A dual motor driver (H-Bridge DC 3 36V 10A Peak 30A Speed Control PWM Module, MOSFET IRF3205) to receive commands and run the motor with a high current. The system's mobility is controlled via Proportional-Integral-Derivative using an ArduPilot control board.

4 Test Results

4.1 Vision based avoidance and tracking

4.1.1 Obstacle Avoidance

For obstacle avoidance, the UAV could find where to move in case of detecting any threat. As shown in Figure 31, the method interprets the red-colored optical flow vectors as a threat and sends the command to move the drone in a specific direction. The desired direction is printed in the terminal window as well. We can see segment 3 is found to be including the most threatening flows and segment 1 is the optimal position for the UAV to move with less power consumption since moving up and down of the UAV requires more energy to be used by the motors.

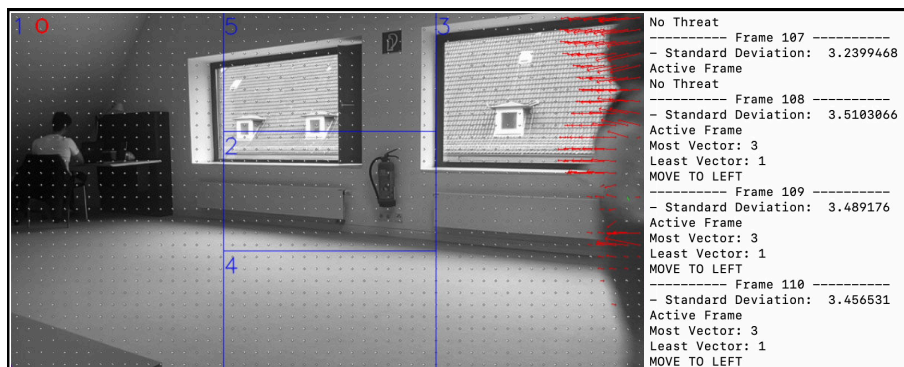


Figure 31: Obstacle Avoiding pose decision test($T_1 = 150$, $T_2 = 3$, $T_3 = 0.1$)

4.1.2 Object Tracking

For object tracking, the method is tested in three situations: when the UAV takes off facing the QR code, back on to the QR code, and without the QR code nearby. In the first situation, the UAV could find the QR code right after the takeoff and directly head toward the code depending on which view segment the bounding box's center aligns. Meanwhile, in the second situation, the UAV could not find the QR in the first takeoff view, thereby rotating itself until it found the QR or fully rotated 360 degrees. In the third situation, the UAV is placed where QR cannot be seen around, and the UAV rotates 360 degrees and then alerts that no object is found to be tracked. Figure 32 shows four frames taken by the UAV's camera during the test on the second situation, which includes all the takeoff, rotation, detection, and tracking performances.

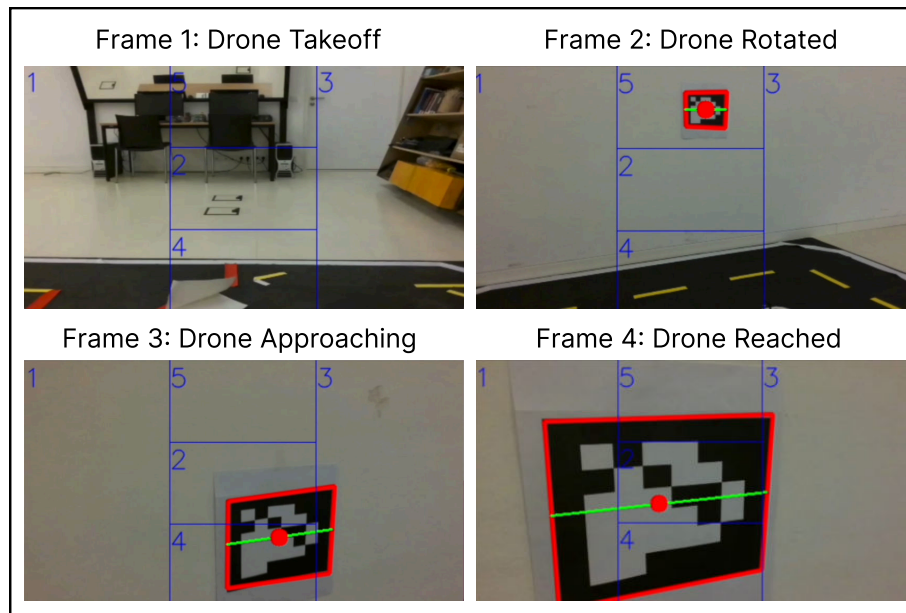
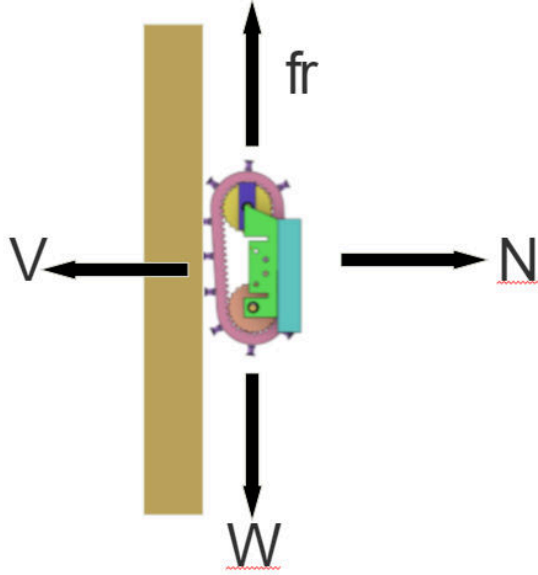


Figure 32: UAV's camera view during QR code tracking performance

4.2 Forces analysis

4.2.1 Body on no load



The Forces acting on the body are:

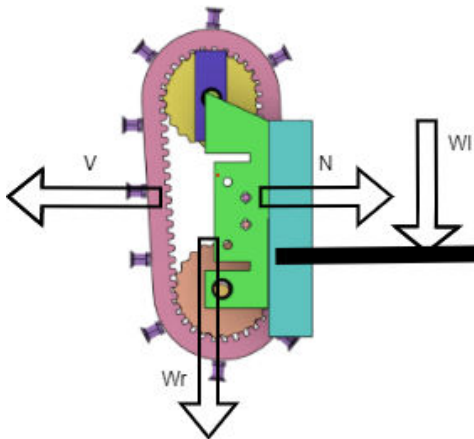
- r : Frictional Force due to the blade.
- N : Normal force.
- V : Adhesion force due to suction cups.
- W : Gravitational Force of the robot.

Figure 34: equations

Figure 33: free body diagram

4.2.2 Arm attached

Our future aim is also to develop a mobile robot capable not only of inspection, but also maintenance and operations on the wind turbine. In this case, an arm should be attached to perform tasks such as cleaning, dust vacuuming and repairing. The weight of the arm should also be taken into consideration when building the adhesion system.



$$\Sigma F_x = 0 \quad (7)$$

$$F_x = fr - W_r - W_l \quad (8)$$

$$\Sigma F_y = 0 \quad (9)$$

$$F_y = N - V \quad (10)$$

Figure 35: free body diagram on load

The free body diagram ?? shows the forces acting on the robot when an arm is attached:

- V : the adhesion force.
- N : the normal force.
- W_R : weight of the robot.
- R : friction force
- W_I : the weight of the inspection tool.
- From the equilibrium equation, we can deduct the maximum weight carried by the adhesion system.
- Equation 7 shows the summation of forces in the x-direction
- equation 8 shows the summation of forces in the y-direction.
- equation 10 is the friction force equation.

4.2.3 Test for passive suction cups

This experiment is designed to confirm that enough pushing force is delivered to the front suction cup during its attachment.

A motor drives a plate with a suction cup attached in the bottom in the vertical direction. The bottom parts contains a force sensor. The motor drives the suction downwards until a maximum pressing force is read by the force sensor, then motor starts driving the suction cups upwards. The force sensor generates the values of the external force applied on the suction cup to the interface, a linear position sensor can generate data about the displacement of the suction cup respectively.

Therefore, we can deduce the relationship between the pressing force and the detaching force of the suction cup, as well as the correlation between the suction cup's displacement and the external force.

By varying the pressing force value, we can conduct series of experiments to calculate the optimal pressing force of the suction cup. [26]

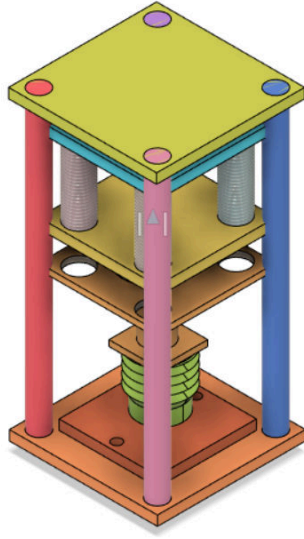


Figure 36: 3D model of test set

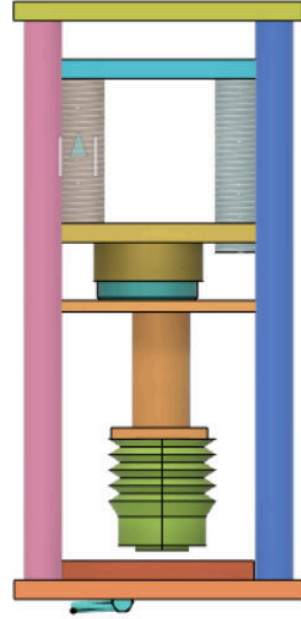


Figure 37: 2D view

5 Conclusions and Future Works

5.1 Obstacle Avoidance and Object Tracking

To achieve the human independence of the wind blade maintenance system, the research proposed the idea of the UAV carrying the mobile robot to the wind blade. Two main objectives, obstacle avoidance and object tracking, are researched in the existence of the UAV's monocular camera image data. The UAV is expected to avoid any unexpected dynamic obstacle by optical flow calculation and analysis and position precisely next to the blade by applying the SIFT method on the QR code and each frame image. According to the testings, the objectives are pretty met yet have further development potential.

In the future, this research on developing a human-independent wind blade maintenance system can be further applied to the maintenance of several other systems such as railroads, factories, and pipelines. With obstacle avoidance as a fundamental feature for safety-ensured flight, the user can set the reference image for the UAV to track to arrive at the specific position of the system. Followingly, the UAV will pause its flight when it is at the goal position and hover. Meanwhile, the deployment system of the UAV will land the mobile robot, which will continue its next phase of contact-based maintenance.

Better disregard of irrelevant optical flow vectors

Still, for obstacle avoidance, simply calculating the STDEV of the view's entire optical flow vectors magnitudes for frame activeness determination is insufficient for specific situations. For example, suppose the UAV is looking at the plain background with some static objects in front of it. In that case, the optical flow vectors derived by the relative velocity tend to occur

on the static objects but not on the background. The major reason is that the static object's pixel features are similar; henceforth, optical flow calculation cannot detect the displacement of the specific point in the background. This situation leads to the case of increasing overall STDEV, which further becomes the factor in interpreting the view as active. The research assumes the UAV's view barely includes sky, ground, and the wind turbines in the far distance, that the suggested challenge might not be so critical. However, for further application of proper obstacle avoidance, a better way of distinguishing the static and dynamic objects will be needed.

As an alternative, further research can instead focus on depth calculation. By achieving the goal of calculating the pixel depth, the UAV can distinguish the threat based on the distance between itself and the obstacle, not by the factor of static or dynamic obstacle. Also, by getting the pixel depth information, the research can aim to calculate 3D optical flow vectors, where the information about whether the obstacle is moving toward the UAV or away from the UAV can be estimated. The problem can be present by observing 2D optical flow vectors, as shown in Figure 13. In Figure 38, the motorcycle moves forward, which is not a threat to the UAV. Still, the 2D optical flow vector analysis says that the wheel creates the threatening optical flow vectors. With depth data, the method can tell that the threatening optical flow vectors created by the motorcycle's wheel will not be affecting the UAV's flight path. Ultimately, the above issue of optical flow vectors created by relative velocity on static objects can also be resolved. The depth information can be provided by the setting of the stereo camera system or the attachment of a distance sensor on the UAV [16]. Thereby, the research leaves pixel depth calculation for further development on the optical flow analysis as future research, which will have a form of the decision tree as shown in Figure 39.



Figure 38: Limited interpretation of 2D optical flow

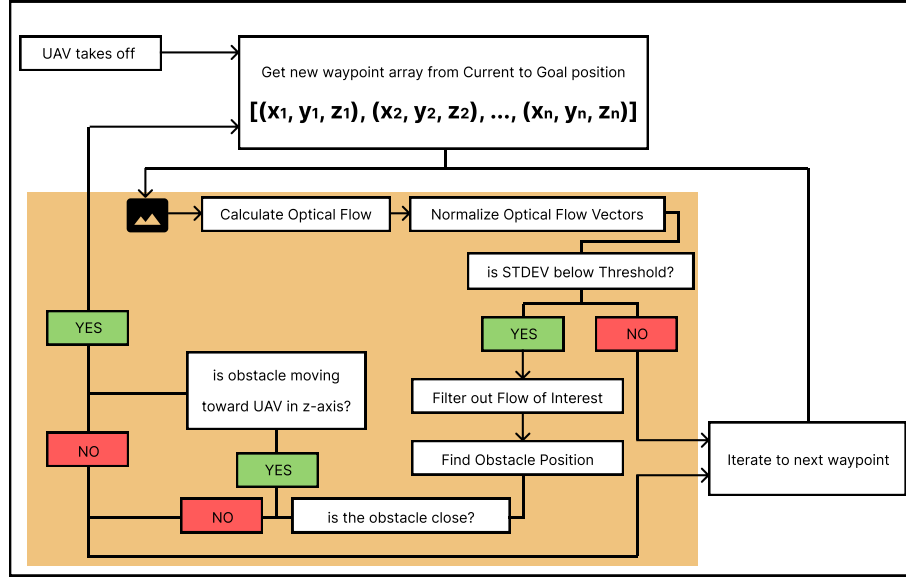


Figure 39: UAV's 3D optical flow based obstacle avoidance system decision tree

Detection of wind turbine hub

The object tracking of the QR code could be done simply with SIFT, one of the classical computer vision techniques. Although, the drawback is that the wind turbine in need of maintenance must have a giant QR code on its hub. No matter whether painting or pasting a sticker, this excessive work is indeed timely and costly expensive. Also, since the environmental harm caused to the QR as time goes by may yield the failure of the QR detection, a frequent check of the QR code's status will also be required. In a nutshell, the research proposes to set another reference object for the UAV to track, the wind turbine's hub.

When we imagine the camera view of the UAV around WP1, the wind blades might not fit into the view because of their massive size. The hub, however, is relatively condensed in size and outstanding in the feature that three wind blades are stretching out of it. Thereby, usage of object detection models such as You Only Look Once (YOLO) trained on the hub image data might help the UAV to detect the hub easily, the case in which excessive work for adding a unique reference object to follow is not required [2]. Therefore, the research proposes using an object detection model trained on hub image data for object detection and tracking for the global use of the system.

Near-Wall Approach of the UAV

One of the expected challenges in the actual industry is the aerodynamic distortions created near the wind blade. When the UAV approaches close to the wind blade, changing aerodynamic force will be created depending on the distance and angle between the UAV and the blade's surface. Referring to the research on the precise positioning and movement of UAVs for near-wall tasks, further research may elaborate on the positioning of the UAV [19] near the wind blade for the precise deployment of the mobile robot.

Threshold optimization

Lastly, the setting of several thresholds(T1, T2, T3) is mandatory for analyzing frame view optical flows. So far, the thresholds are defined manually to fit into the local test cases. To

optimize those values to be the global thresholds later, the research proposes to apply the gradient descent method to find the corresponding values [13]. For example, T1 and T2 are responsible factors for determining the frame's activeness. Assume we have 100 sample images to test the thresholds. We collect data points (T1, T2, cnt_loss) manually. At this moment, cnt_loss stands for (number of ground truths = 100) - (number of correct results using T1 and T2 as thresholds). After then, we can interpolate the data points to find the pattern function. Finally, by applying the gradient descent method, we will be able to find the point (T1, T2) with the least cnt_loss. The research leaves this stage of collecting ground truth data and applying the gradient descent method on the thresholds to determine the activeness of the frame for future research.

5.2 Financial consequences overview

The purpose of this research is to develop solutions for energy firms. Because wind turbines maintenance and inspection may be risky and costly for humans, a wall climbing robot is not only capable of operating in hazardous areas, but can also result in significant cost savings for businesses when compared to hiring qualified personnel.

N	Unit Cost	Working time	Total cost
Labor force (Five person)	25-40 Euros per hour	4-6 hours	500-800 Euros
Robot	One time purchase of 400	Continuous Power Supply	400

Table 4: Comparative cost for wind turbine inspection

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